# PREETI RAMESH

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## EDUCATION

**University of Southern California** — Los Angeles, CA, USA
**M.S in Electrical and Computer Engineering (Machine Learning and Data Science) | GPA: 3.92/4** — Aug 2025-May 2027

**National Institute of Technology, Tiruchirappalli** — Tamil Nadu, India
**B. Tech in Electronics and Communication Engineering | CGPA: 3.51/4** — Jul 2021-May 2025

## PROFESSIONAL EXPERIENCE

**Becton, Dickinson and Company (BD)** — Aug 2026-Present
**Applied Machine Learning Co-op Intern** — Los Angeles, CA, USA

- Investigating cuffless blood pressure estimation using PPG foundation models and clinical waveform datasets, including representation learning, model benchmarking, and calibration strategy evaluation for wearable health monitoring.

**Children's Hospital Los Angeles (CHLA)** — Jun 2026-Aug 2026
**Clinical Research Scholar (Levitt Lab)** — Los Angeles, CA, USA

- Developed reproducible Python pipelines integrating longitudinal REDCap data from the Family First longitudinal mother–infant cohort, automating cohort extraction and generation of analysis-ready datasets to support biomarker discovery related to early-life adversity.
- Performed hypothesis-driven statistical analyses investigating the biological embedding of early-life adversity by examining associations between questionnaire-derived adversity measures (ACEs, PEARLS), autonomic regulation (Heart Rate Variability [HRV], Respiratory Sinus Arrhythmia [RSA]), oxidative stress biomarkers (F2-isoprostanes), and Bayley-IV developmental outcomes.
- Designed and implemented an interactive Python graphic user interface (GUI) for longitudinal clinical data exploration and visualization, enabling researchers to inspect participant-level trajectories, compare candidate biomarkers, and accelerate clinical research workflows.

**Argonne National Laboratory** — Jan 2025-Apr 2025
**Machine Learning Intern (Transportation and Power Systems)** — Chicago, IL, USA

- Developed a virtual electronic control unit (ECU) system for Connected & Autonomous vehicles (CAVs) using Python to capture, emulate, and replay Controller Area Network (CAN) diagnostic communication, enabling offline algorithm validation and system behavior analysis.
- Created an interactive graphic user interface (GUI) enabling low-level message inspection, byte-wise response modification, and deterministic request-response replay, improving the interpretability of embedded communication pipelines.
- Investigated transmission latency and protocol inconsistencies while using the ESP32 microcontroller setup, ensuring high-throughput data integrity and robust system performance. (Poster)

**Agency for Science, Technology and Research** — May 2024-Aug 2024
**Data Science Intern (Healthcare & Medtech Division)** — One North, Singapore

- Conducted extensive preprocessing on provided EMG (Electromyography) data for **RESTORES** (**REST**oration **O**f **R**ehabilitative function with **E**pidural spinal **S**timulation) project for patients with chronic spinal cord injury.
- Evaluated multiple techniques such as filters, wavelet decomposition, noise and artifact handling to establish a universal benchmark preprocessing method tailored to given datasets.
- Constructed algorithms in MATLAB to examine EMG (Electromyography) signals and study how different stimulation parameters (frequency and current) affect patient movement, helping optimize treatment protocols and improve patient outcomes. (Project Presentation)

## PUBLICATIONS

Sharan Roy, Shreya Manish Band, **Preeti Ramesh**, Punitkumar Bhavsar, Ar Jac Fredo, Deepesh Kumar. "Advancing Emotion Detection through Dual-Channel EEG and Peripheral Physiological Signal Integration," 2024 International Conference on Modeling, Simulation & Intelligent Computing (MoSICom), Dubai, United Arab Emirates, 2024, pp. 105-109, doi: 10.1109/MoSICom63082.2024.10881605

## PROJECTS

**University of Southern California** — Los Angeles, CA, USA
**Graduate Researcher (Biomedical Imaging Group - BIG Lab)** — Jan 2026 - Present

- Developing a cross-platform Brainstorm (MATLAB)-to-Python EEG processing pipeline for epilepsy research, transforming heterogeneous clinical EEG recordings into standardized, machine learning-ready datasets through signal preprocessing, channel mapping, montage harmonization, and metadata organization.
- Designing and evaluating deep learning algorithms for automated epileptiform spike detection, benchmarking model performance on expert-annotated EEG recordings to enable scalable clinical EEG/sEEG analysis and reduce clinician review time.

**Volunteer Researcher**: **(Khan Lab)** — Dec 2025 - Present

- Developing a generalized wearable stress-detection framework for a multimodal wearable sensing platform using EDA and PPG physiological signals, integrating physiological signal preprocessing, HRV analysis, automated feature extraction, and interpretable feature engineering to enable robust stress detection across diverse physiological stress paradigms, with the Cold Pressor Test (CPT) serving as the baseline validation model.
- Built and benchmarked subject-independent machine learning workflows using LOSO cross-validation, Sequential Feature Selection (SFS), and multimodal classification, achieving 82.2% accuracy (80.6% balanced accuracy) on the initial Cold Pressor Test (CPT) benchmark and establishing a reproducible pipeline for validation across multiple stress paradigms.

**Indian Institute of Technology (BHU)** — May 2023 - Jul 2023
**Research Intern (School of Biomedical Engineering)** — Uttar Pradesh, India

- Developed and evaluated classical machine learning techniques along with deep learning models, including 1D CNN (One-Dimensional Convolutional Neural Network) and SWT CNN (Single Wavelet Transform Convolutional Neural Network), for emotion recognition from electrodermal activity (EDA) physiological time-series using PyTorch and Keras.
- Created a robust classification system to accurately categorize emotions into valence and arousal dimensions, providing insights into individuals' emotional states
- Directed testing and evaluation to validate accuracy and efficacy of emotion analysis framework, yielding significant findings.

**National Institute of Technology, Tiruchirappalli** — Dec 2022 - Jan 2023
**Research Intern (Department of Instrumentation & Control Engineering)** — Tamil Nadu**,** India

- Assisted in creating a low-cost electronic stethoscope and developed digital signal processing pipelines for reliable respiratory sound acquisition and analysis.

## SKILLS

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| **Programming Languages:** | Python, MATLAB |
| **Frameworks**: | PyTorch, TensorFlow/Keras, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Firebase, Neurokit2, MNE-Python, Brainstorm, Jupyter Notebook, VS Code |
| **Areas of Expertise:** | Artificial Intelligence & Machine Learning for Healthcare, Neural Networks & Data Science, Deep Learning, Biomedical Signal Processing, Time-Series Analysis, Predictive Modeling, Statistical Analysis, Multimodal Data Analysis |
| **Biomedical Signals:** | EEG (Electroencephalogram), MEG (Magnetoencephalography), EMG (Electromyography), ECG (Electrocardiogram), PPG (Photoplethysmography), HRV (Heart Rate Variability), EDA (Electrodermal Activity), RSA (Respiratory Sinus Arrhythmia), Respiratory Audio (Electronic Stethoscope), Physiological Signal Processing, Neurotechnology |
| **Instrumentation:** | Biomedical Instrumentation & Data Acquisition, Embedded Systems & Communication, Real-Time Systems & Monitoring, Protocol Testing and Debugging, CAN Bus, ESP32 |